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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 29th January Evening Shift

JEE Main Question Paper with Answer Key

29 Jan 2023	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

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Out of syllabus	20 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The decrease in first ionization enthalpy from B to Al is much larger than that from Al to Ga.

Statement II : The d orbitals in Ga are completely filled.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Statement I is correct but statement II is incorrect

(B) Statement I is incorrect but statement II is correct

(C) Both the statements I and II are correct

(D) Both the statements I and II are incorrect

Q2. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Nickel is being used as the catalyst for producing syn gas and edible fats.

Statement II : Silicon forms both electron rich and electron deficient hydrides.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but statement II is incorrect

(B) Both the statements I and II are correct

(C) Both the statements I and II are incorrect

(D) Statement I is incorrect but statement II is correct

Q3. MCQ (Single Correct Answer)

Open solution

Which of the following relations are correct?

(A) $\Delta U = q + p\Delta V$

(B) $\Delta G = \Delta H - T\Delta S$

(C) $\Delta S = \frac{q_{rev}}{T}$

(D) $\Delta H = \Delta U - \Delta nRT$

Choose the most appropriate answer from the options given below :

(A) A and B only (B) B and C only (C) C and D only (D) B and D only

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Q4. MCQ (Single Correct Answer)

A solution of CrO_5 in amyl alcohol has a _____ colour.

(A) Yellow (B) Orange-Red (C) Green (D) Blue

Open solution

Q5. MCQ (Single Correct Answer)

According to MO theory the bond orders for O_2^{2-}, CO and NO^+ respectively, are

(A) 1, 3 and 3 (B) 2, 3 and 3 (C) 1, 2 and 3 (D) 1, 3 and 2

Open solution

Q6. MCQ (Single Correct Answer)

When a hydrocarbon A undergoes combustion in the presence of air, it requires 9.5 equivalents of oxygen and produces 3 equivalents of water. What is the molecular formula of A?

(A) C_9H_6 (B) C_6H_6 (C) C_8H_6 (D) C_9H_9

Open solution

Q7. MCQ (Single Correct Answer)

The set of correct statements is :

(i) Manganese exhibits +7 oxidation state in its oxide.

(ii) Ruthenium and Osmium exhibit +8 oxidation state in their oxides.

(iii) Sc shows +4 oxidation state which is oxidizing in nature.

(iv) Cr shows oxidising nature in +6 oxidation state.

(A) (ii), (iii) and (iv) (B) (ii) and (iii)

(C) (i) and (iii) (D) (i), (ii) and (iv)

Open solution

Q8. MCQ (Single Correct Answer)

Correct order of spin only magnetic moment of the following complex ions is :

(Given At.no. Fe : 26, Co : 27)

(A) $[CoF_6]^{3-} > [FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$ (B) $[FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-}$

(C) $[FeF_6]^{3-} > [CoF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$ (D) $[Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-} > [FeF_6]^{3-}$

Open solution

Q9. MCQ (Single Correct Answer)

Reaction of propanamide with $Br_2/KOH(aq)$ produces :

(A) Propanenitrile (B) Propylamine (C) Ethylnitrile (D) Ethylamine

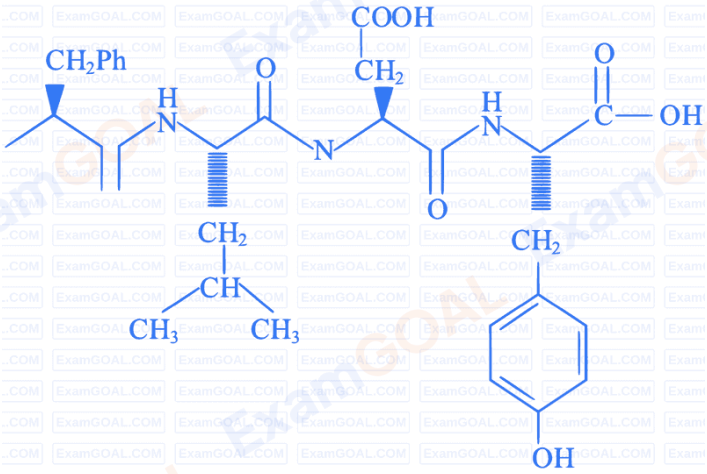
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Q10. MCQ (Single Correct Answer)

Open solution

Following tetrapeptide can be represented as



(F, L, D, Y, I, Q, P are one letter codes for amino acids)

(A) YQLF (B) FIQY (C) PLDY (D) FLDY

Q11. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List I		List II
A.	van't Hoff factor, i	I.	Cryoscopic constant
B.	k_f	II.	Isotonic solutions
C.	Solutions with same osmotic pressure	III.	$\frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$
D.	Azeotropes	IV.	Solutions with same composition of vapour above it

Choose the correct answer from the options given below :

(A) A-III, B-I, C-IV, D-II (B) A-III, B-I, C-II, D-IV

(C) A-III, B-II, C-I, D-IV (D) A-I, B-III, C-II, D-IV

Q12. MCQ (Single Correct Answer)

Open solution

The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement)

(A) 2-Bromo-3,3-dimethylpentane (B) 2-Bromopropane

(C) 1-Bromo-2-methylbutane (D) 2-Bromopentane

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Q13. MCQ (Single Correct Answer)

Open solution

An indicator 'X' is used for studying the effect of variation in concentration of iodide on the rate of reaction of iodide ion with H_2O_2 at room temp. The indicator 'X' forms blue coloured complex with compound 'A' present in the solution. The indicator 'X' and compound 'A' respectively are :

(A) Methyl orange and H_2O_2

(B) Methyl orange and iodine

(C) Starch and H_2O_2

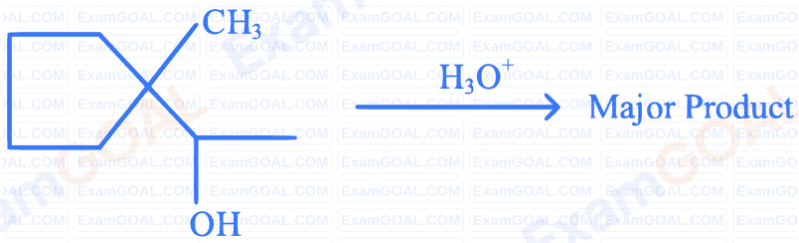
(D) Starch and iodine

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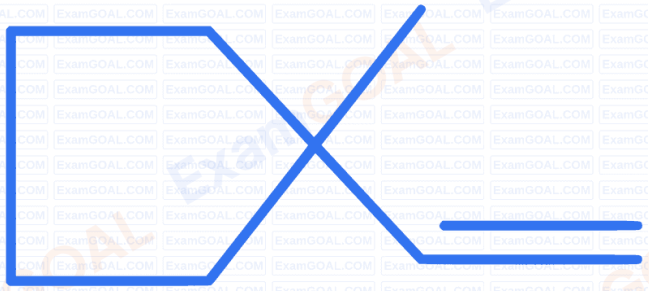
Q14. MCQ (Single Correct Answer)

Open solution

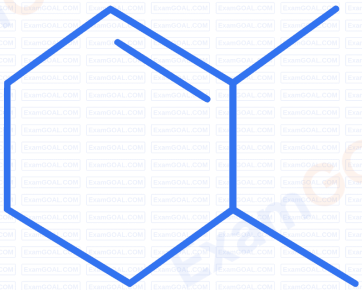
Find out the major product for the following reaction.



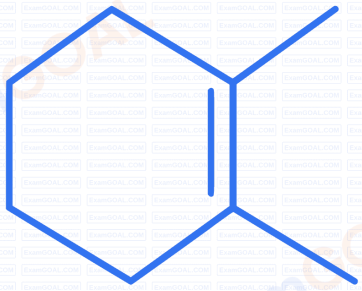
(A)



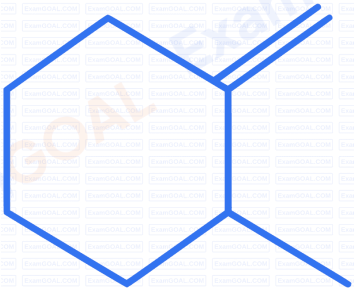
(B)



(C)



(D)

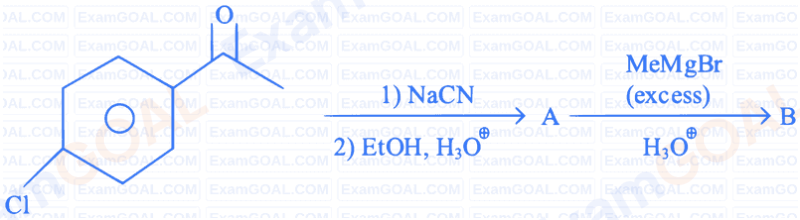


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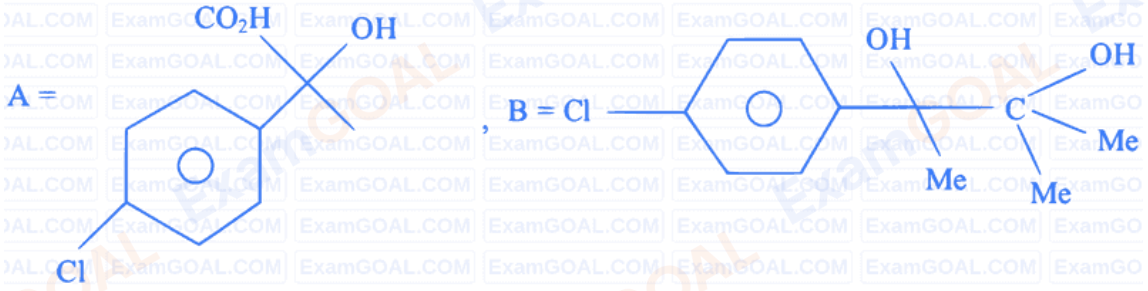
Q15. MCQ (Single Correct Answer)

Open solution

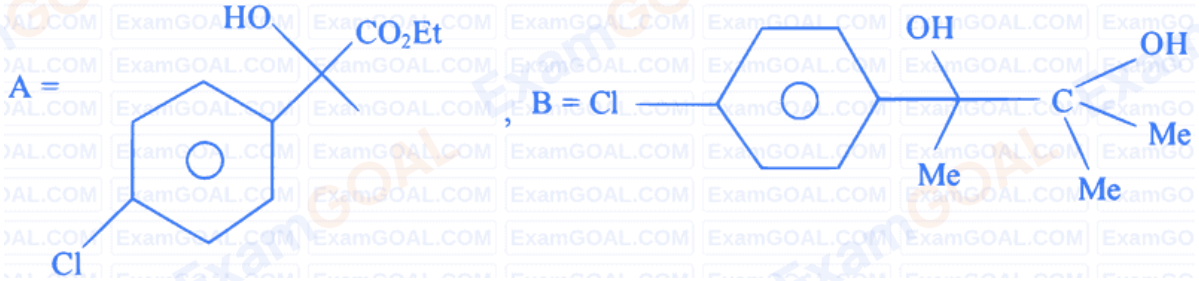
Find out the major products from the following reaction sequence.



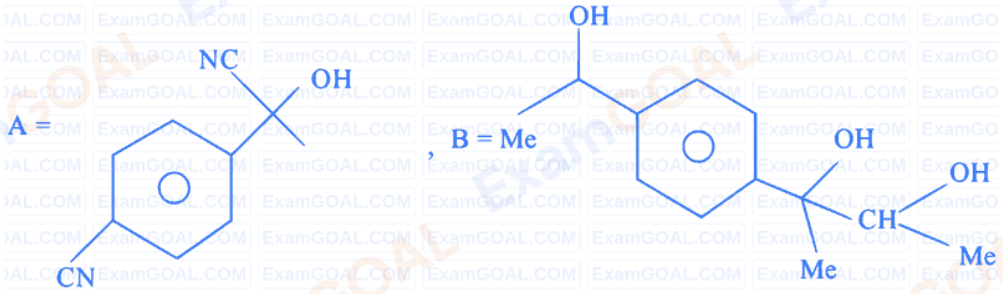
(A)



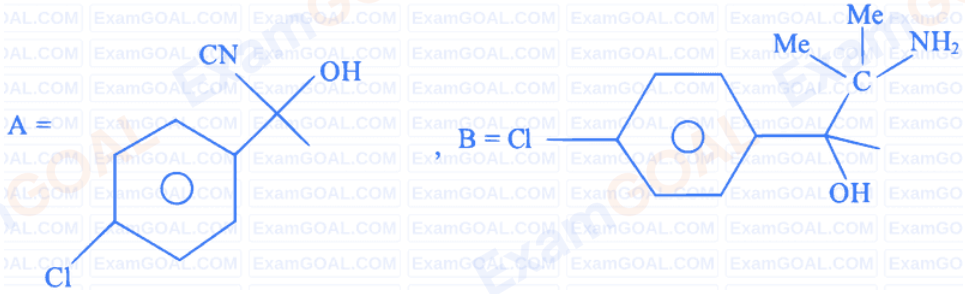
(B)



(C)



(D)



Q16. Numerical

Open solution

The volume of HCl, containing 73 g L^{-1}, required to completely neutralise NaOH obtained by reacting 0.69 g of metallic sodium with water, is _____ mL. (Nearest Integer)

(Given : molar masses of Na, Cl, O, H, are 23, 35.5, 16 and 1 g mol^{-1} respectively.)

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Q17. Numerical

The equilibrium constant for the reaction

$Zn(s) + Sn^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Sn(s)$ is 1×10^{20} at 298 K. The magnitude of standard electrode potential of Sn/Sn^{2+} if $E^{\Theta}_{Zn^{2+}/Zn} = -0.76$ V is _____ $\times 10^{-2}$ V. (Nearest integer)

Given : $\frac{2.303RT}{F} = 0.059$ V

Open solution

Q18. Numerical

At 298 K

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $K_1 = 4 \times 10^5$

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $K_2 = 1.6 \times 10^{12}$

$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g)$, $K_3 = 1.0 \times 10^{-13}$

Based on above equilibria, then equilibrium constant of the reaction,
$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$ is _____ $\times 10^{-33}$ (Nearest integer).

Open solution

Q19. Numerical

The denticity of the ligand present in the Fehling's reagent is _____.

Open solution

Q20. Numerical

When 0.01 mol of an organic compound containing 60% carbon was burnt completely, 4.4 g of CO_2 was produced. The molar mass of compound is _____ g mol^{-1} (Nearest integer).

Open solution

Q21. Numerical

Total number of acidic oxides among

$N_2O_3, NO_2, N_2O, Cl_2O_7, SO_2, CO, CaO, Na_2O$ and NO is _____.

Open solution

Q22. Numerical

Assume that the radius of the first Bohr orbit of hydrogen atom is 0.6 $\mathring{A}$. The radius of the third Bohr orbit of He^+ is _____ picometer. (Nearest Integer)

Open solution

Q23. Numerical

For conversion of compound A $\rightarrow$ B, the rate constant of the reaction was found to be 4.6×10^{-5} L mol^{-1} s^{-1}. The order of the reaction is _____.

Open solution

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Mathematics 21 questions

Q24. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution of the differential equation $x\log_e x\frac{dy}{dx} + y = x^2\log_e x, (x > 1)$. If $y(2) = 2$, then $y(e)$ is equal to

(A) $\frac{1+e^2}{2}$ (B) $\frac{1+e^2}{4}$ (C) $\frac{2+e^2}{2}$ (D) $\frac{4+e^2}{4}$

Open solution

Q25. MCQ (Single Correct Answer)

Let R be a relation defined on $\mathbb{N}$ as $a\mathrm{R}b$ if $2a + 3b$ is a multiple of $5, a, b \in \mathbb{N}$. Then R is

(A) an equivalence relation (B) non reflexive

(C) symmetric but not transitive (D) transitive but not symmetric

Open solution

Q26. MCQ (Single Correct Answer)

The letters of the word OUGHT are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word TOUGH is :

(A) 79 (B) 84 (C) 89 (D) 86

Open solution

Q27. MCQ (Single Correct Answer)

If $\vec{a} = \hat{i} + 2\hat{k}, \vec{b} = \hat{i} + \hat{j} + \hat{k}, \vec{c} = 7\hat{i} - 3\hat{j} + 4\hat{k}, \vec{r} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$ and $\vec{r}.\vec{a} = 0$. Then $\vec{r}.\vec{c}$ is equal to :

(A) 36 (B) 30 (C) 34 (D) 32

Open solution

Q28. MCQ (Single Correct Answer)

The number of 3 digit numbers, that are divisible by either 3 or 4 but not divisible by 48, is :

(A) 400 (B) 472 (C) 507 (D) 432

Open solution

Q29. MCQ (Single Correct Answer)

Let $\mathrm{S} = \{w_1, w_2, \ldots\ldots\}$ be the sample space associated to a random experiment. Let $P(w_n) = \frac{P(w_{n-1})}{2}, n \geq 2$. Let $A = \{2k + 3l : k, l \in N\}$ and $B = \{w_n : n \in A\}$. Then P(B) is equal to :

(A) $\frac{3}{32}$ (B) $\frac{1}{32}$ (C) $\frac{1}{16}$ (D) $\frac{3}{64}$

Open solution

Q30. MCQ (Single Correct Answer)

Consider a function $f : \mathbb{N} \to \mathbb{R}$, satisfying $f(1) + 2f(2) + 3f(3) + \ldots + xf(x) = x(x+1)f(x); x \geq 2$ with $f(1) = 1$. Then $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to

(A) 8000 (B) 8400 (C) 8100 (D) 8200

Open solution

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Q31. MCQ (Single Correct Answer)

The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ has a real solution x, is :

(A) $[-2, -1]$ (B) $\left[-\frac{3}{2}, -1\right]$

(C) $\left[-2, -\frac{3}{2}\right]$ (D) $\left[-1, -\frac{1}{2}\right]$

Open solution

Q32. MCQ (Single Correct Answer)

The area of the region $A = \{(x, y) : |\cos x - \sin x| \le y \le \sin x, 0 \le x \le \frac{\pi}{2}\}$ is

(A) $\sqrt{5} + 2\sqrt{2} - 4.5$ (B) $1 - \frac{3}{\sqrt{2}} + \frac{4}{\sqrt{5}}$

(C) $\sqrt{5} - 2\sqrt{2} + 1$ (D) $\frac{3}{\sqrt{5}} - \frac{3}{\sqrt{2}} + 1$

Open solution

Q33. MCQ (Single Correct Answer)

Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = 3\hat{i} - 4\hat{j} + 5\hat{k}$. If $\vec{c}$ is a vector such that $\vec{c}.\left(\vec{a} \times \vec{b}\right) + 25 = 0, \vec{c}.(\hat{i} + \hat{j} + \hat{k}) = 4$, and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals :

(A) $\frac{3}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{5}$ (D) $\frac{5}{\sqrt{2}}$

Open solution

Q34. MCQ (Single Correct Answer)

The value of the integral $\int_1^2 \left(\frac{t^4+1}{t^6+1}\right) dt$ is

(A) $\tan^{-1}\frac{1}{2} - \frac{1}{3}\tan^{-1}8 + \frac{\pi}{3}$ (B) $\tan^{-1}2 - \frac{1}{3}\tan^{-1}8 + \frac{\pi}{3}$

(C) $\tan^{-1}2 + \frac{1}{3}\tan^{-1}8 - \frac{\pi}{3}$ (D) $\tan^{-1}\frac{1}{2} + \frac{1}{3}\tan^{-1}8 - \frac{\pi}{3}$

Open solution

Q35. MCQ (Single Correct Answer)

Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1 + x)^{99}$. Let a be the middle term in the expansion of $\left(2 + \frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{^{200}C_{99}K}{a} = \frac{2^l m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to

(A) (50, 101) (B) (50, 51) (C) (51, 101) (D) (51, 99)

Open solution

Q36. MCQ (Single Correct Answer)

The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1}x}{x} dx$ is equal to :

(A) $\frac{\pi}{2}\log_e 2$ (B) $\frac{\pi}{4}\log_e 2$ (C) $\frac{1}{2}\log_e 2$ (D) $\pi\log_e 2$

Open solution

Q37. MCQ (Single Correct Answer)

The shortest distance between the lines $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ and $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ is :

(A) $2\sqrt{3}$ (B) $3\sqrt{3}$ (C) $4\sqrt{3}$ (D) $5\sqrt{3}$

Open solution

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Q38. MCQ (Single Correct Answer)

Let f and g be the twice differentiable functions on $\mathbb{R}$ such that

$f''(x) = g''(x) + 6x$

$f'(1) = 4g'(1) - 3 = 9$

$f(2) = 3g(2) = 12.$

Then which of the following is NOT true?

(A) $g(-2) - f(-2) = 20$

(B) There exists $x_0 \in (1, 3/2)$ such that $f(x_0) = g(x_0)$

(C) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

(D) If $-1 < x < 2$, then $|f(x) - g(x)| < 8$

Open solution

Q39. Numerical

The total number of 4-digit numbers whose greatest common divisor with 54 is 2, is _____.

Open solution

Q40. Numerical

Let $X = \{11, 12, 13, \ldots, 40, 41\}$ and $Y = \{61, 62, 63, \ldots, 90, 91\}$ be the two sets of observations. If $\overline{x}$ and $\overline{y}$ are their respective means and σ^2 is the variance of all the observations in $X \cup Y$, then $\left|\overline{x} + \overline{y} - \sigma^2\right|$ is equal to _____.

Open solution

Q41. Numerical

Let $\alpha = 8 - 14i$, $A = \left\{z \in \mathbb{C} : \frac{\alpha z - \overline{\alpha z}}{z^2 - (\overline{z})^2 - 112i} = 1\right\}$ and $B = \{z \in \mathbb{C} : |z + 3i| = 4\}$. Then $\sum\limits_{z \in A \cap B} (\mathrm{Re}z - \mathrm{Im}z)$ is equal to _____.

Open solution

Q42. Numerical

Let $\alpha_1, \alpha_2, \ldots, \alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and $|\alpha_1| \geq |\alpha_2| \geq \ldots \geq |\alpha_7|$. Then $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____.

Open solution

Q43. Numerical

Let $\{a_k\}$ and $\{b_k\}, k \in N$, be two G.P.s with common ratios r_1 and r_2 respectively such that $a_1 = b_1 = 4$ and $r_1 < r_2$. Let $c_k = a_k + b_k, k \in N$. If $c_2 = 5$ and $c_3 = \frac{13}{4}$ then $\sum\limits_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$ is equal to _____.

Open solution

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Q44. Numerical

Let A be a symmetric matrix such that $|A| = 2$ and $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$. If the sum of the diagonal elements of A is s, then $\frac{\beta s}{\alpha^2}$ is equal to _____.

Open solution

Physics 26 questions

Q45. MCQ (Single Correct Answer)

The ratio of de-Broglie wavelength of an α particle and a proton accelerated from rest by the same potential is $\frac{1}{\sqrt{m}}$, the value of m is -

(A) 2 (B) 16 (C) 8 (D) 4

Open solution

Q46. MCQ (Single Correct Answer)

A force acts for 20 s on a body of mass 20 kg, starting from rest, after which the force ceases and then body describes 50 m in the next 10 s. The value of force will be:

(A) 40 N (B) 20 N (C) 5 N (D) 10 N

Open solution

Q47. MCQ (Single Correct Answer)

The equation of a circle is given by $x^2 + y^2 = a^2$, where a is the radius. If the equation is modified to change the origin other than (0, 0), then find out the correct dimensions of A and B in a new equation : $(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2$. The dimensions of t is given as $[\mathrm{T}^{-1}]$.

(A) $\mathrm{A} = [\mathrm{L}^{-1}\mathrm{T}^{-1}], \mathrm{B} = [\mathrm{LT}^{-1}]$ (B) $\mathrm{A} = [\mathrm{L}^{-1}\mathrm{T}^{-1}], \mathrm{B} = [\mathrm{LT}]$

(C) $\mathrm{A} = [\mathrm{LT}], \mathrm{B} = [\mathrm{L}^{-1}\mathrm{T}^{-1}]$ (D) $\mathrm{A} = [\mathrm{L}^{-1}\mathrm{T}], \mathrm{B} = [\mathrm{LT}^{-1}]$

Open solution

Q48. MCQ (Single Correct Answer)

A square loop of area 25 cm^2 has a resistance of 10 Ω. The loop is placed in uniform magnetic field of magnitude 40.0 T. The plane of loop is perpendicular to the magnetic field. The work done in pulling the loop out of the magnetic field slowly and uniformly in 1.0 sec, will be

(A) 1.0×10^{-3} J (B) 5×10^{-3} J (C) 2.5×10^{-3} J (D) 1.0×10^{-4} J

Open solution

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Q49. MCQ (Single Correct Answer)

Open solution

Heat energy of 184 kJ is given to ice of mass 600 g at $-12^{\circ}\mathrm{C}$. Specific heat of ice is $2222.3\ \mathrm{J\ kg^{-1^{\circ}}\ C^{-1}}$ and latent heat of ice is 336 $\mathrm{kJ\ kg^{-1}}$

A. Final temperature of system will be 0°C.

B. Final temperature of the system will be greater than 0°C.

C. The final system will have a mixture of ice and water in the ratio of 5 : 1.

D. The final system will have a mixture of ice and water in the ratio of 1 : 5.

E. The final system will have water only.

Choose the correct answer from the options given below :

(A) A and E only (B) B and D only (C) A and C only (D) A and D only

Q50. MCQ (Single Correct Answer)

Open solution

Identify the correct statements from the following :

A. Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative.

B. Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative.

C. Work done by friction on a body sliding down an inclined plane is positive.

D. Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity is zero.

E. Work done by the air resistance on an oscillating pendulum is negative.

Choose the correct answer from the options given below :

(A) A and C only (B) B and D only (C) B, D and E only (D) B and E only

Q51. MCQ (Single Correct Answer)

Open solution

A scientist is observing a bacteria through a compound microscope. For better analysis and to improve its resolving power he should. (Select the best option)

(A) Decrease the focal length of the eye piece.

(B) Increase the wavelength of the light

(C) Increase the refractive index of the medium between the object and objective lens

(D) Decrease the diameter of the objective lens

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Q52. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Electromagnetic waves are not deflected by electric and magnetic field.

Statement II : The amplitude of electric field and the magnetic field in electromagnetic waves are related to each other as $E_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} B_0$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true and Statement II is false

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

Q53. MCQ (Single Correct Answer)

Open solution

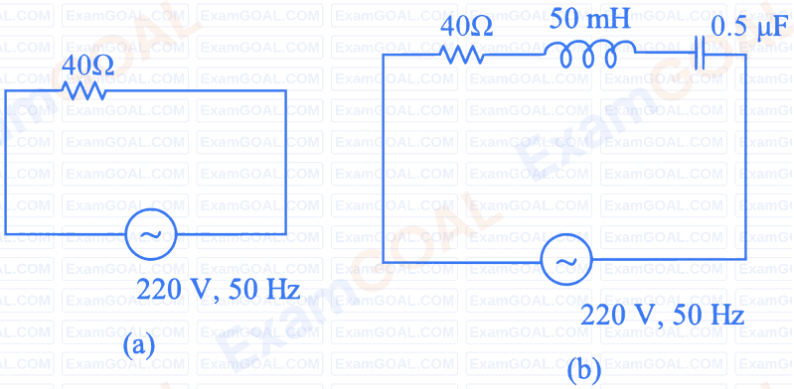
The time taken by an object to slide down 45° rough inclined plane is n times as it takes to slide down a perfectly smooth 45° incline plane. The coefficient of kinetic friction between the object and the incline plane is :

(A) $1 - \frac{1}{n^2}$ (B) $1 + \frac{1}{n^2}$ (C) $\sqrt{1 - \frac{1}{n^2}}$ (D) $\sqrt{\frac{1}{1-n^2}}$

Q54. MCQ (Single Correct Answer)

Open solution

For the given figures, choose the correct options :



(A) The rms current in circuit (b) can be larger than that in (a)

(B) The rms current in figure (a) is always equal to that in figure (b)

(C) The rms current in circuit (b) can never be larger than that in (a)

(D) At resonance, current in (b) is less than that in (a)

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Q55. MCQ (Single Correct Answer)

An object moves at a constant speed along a circular path in a horizontal plane with center at the origin. When the object is at $x = +2\,\text{m}$, its velocity is $-4\hat{\text{j}}$ m/s. The object's velocity (v) and acceleration (a) at $x = -2\,\text{m}$ will be

Open solution

(A) $v = 4\hat{\text{i}}\,\text{m/s}, a = 8\hat{\text{j}}\,\text{m/s}^2$

(B) $v = 4\hat{\text{j}}\,\text{m/s}, a = 8\hat{\text{i}}\,\text{m/s}^2$

(C) $v = -4\hat{\text{i}}\,\text{m/s}, a = -8\hat{\text{j}}\,\text{m/s}^2$

(D) $v = -4\hat{\text{j}}\,\text{m/s}, a = 8\hat{\text{i}}\,\text{m/s}^2$

Q56. MCQ (Single Correct Answer)

The time period of a satellite of earth is 24 hours. If the separation between the earth and the satellite is decreased to one fourth of the previous value, then its new time period will become.

Open solution

(A) 12 hours (B) 3 hours (C) 6 hours (D) 4 hours

Q57. MCQ (Single Correct Answer)

The electric current in a circular coil of four turns produces a magnetic induction 32 T at its centre. The coil is unwound and is rewound into a circular coil of single turn, the magnetic induction at the centre of the coil by the same current will be :

Open solution

(A) 2 T (B) 4 T (C) 8 T (D) 16 T

Q58. MCQ (Single Correct Answer)

A fully loaded boeing aircraft has a mass of 5.4×10^5 kg. Its total wing area is 500 m^2. It is in level flight with a speed of 1080 km/h. If the density of air ρ is 1.2 kg m^{-3}, the fractional increase in the speed of the air on the upper surface of the wing relative to the lower surface in percentage will be. $(\text{g} = 10\,\text{m/s}^2)$

Open solution

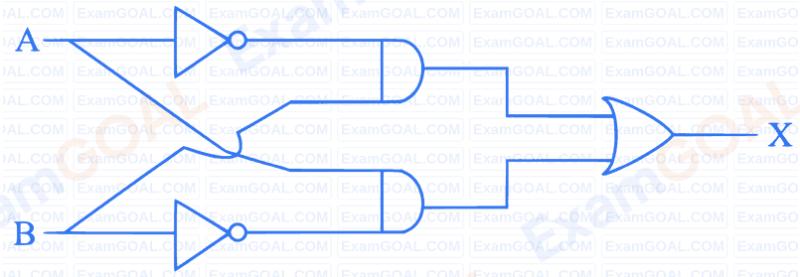
(A) 16 (B) 8 (C) 6 (D) 10

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Q59. MCQ (Single Correct Answer)

Open solution

For the given logic gates combination, the correct truth table will be



(A)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

(B)

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

(C)

A	B	X
0	0	1
0	1	0
1	0	1
1	1	0

(D)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

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Q60. MCQ (Single Correct Answer)

Open solution

At 300 K, the rms speed of oxygen molecules is $\sqrt{\frac{\alpha+5}{\alpha}}$ times to that of its average speed in the gas. Then, the value of α will be

(used $\pi = \frac{22}{7}$)

(A) 27 (B) 28 (C) 24 (D) 32

Q61. MCQ (Single Correct Answer)

Open solution

A point charge 2×10^{-2} C is moved from P to S in a uniform electric field of $30\ \mathrm{NC}^{-1}$ directed along positive x-axis. If coordinates of P and S are (1, 2, 0) m and (0, 0, 0) m respectively, the work done by electric field will be

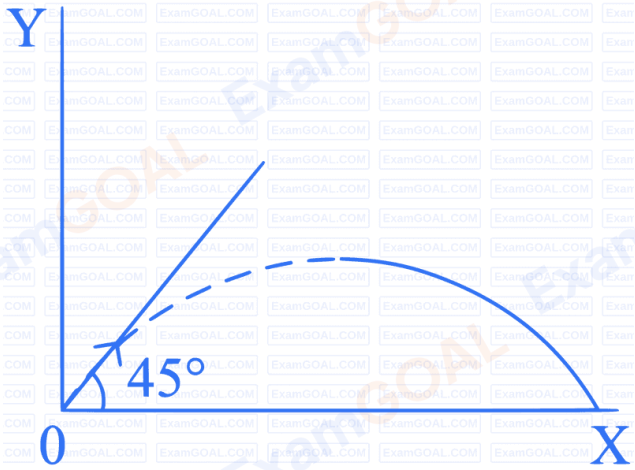
(A) 600 mJ (B) -1200 mJ (C) 1200 mJ (D) -600 mJ

Q62. Numerical

Open solution

A particle of mass 100 g is projected at time t = 0 with a speed 20 ms^{-1} at an angle 45° to the horizontal as given in the figure. The magnitude of the angular momentum of the particle about the starting point at time t = 2s is found to be $\sqrt{\mathrm{K}}\ \mathrm{kg\ m^2/s}$. The value of K is _____.

(Take g = 10 ms^{-2})



Q63. Numerical

Open solution

In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab as 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10} \times 10^{-3}$, where x is _____

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Q64. Numerical

An inductor of inductance 2 μH is connected in series with a resistance, a variable capacitor and an AC source of frequency 7 kHz. The value of capacitance for which maximum current is drawn into the circuit $\frac{1}{x}$F, where the value of x is _____.

(Take $\pi = \frac{22}{7}$)

Open solution

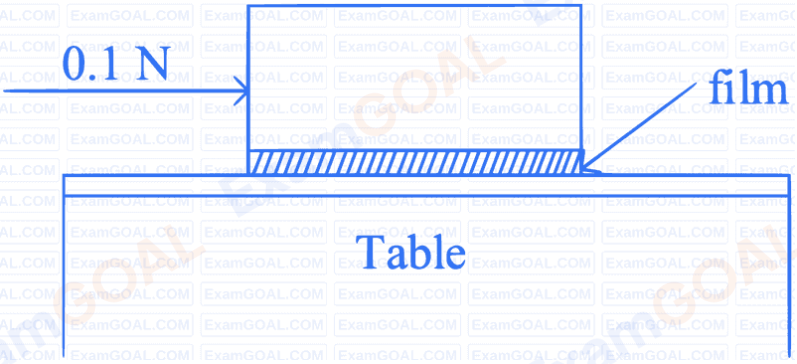
Q65. Numerical

A particle of mass 250 g executes a simple harmonic motion under a periodic force $\mathrm{F} = (-25\,x)\mathrm{N}$. The particle attains a maximum speed of 4 m/s during its oscillation. The amplitude of the motion is _____ cm.

Open solution

Q66. Numerical

A metal block of base area 0.20 m^2 is placed on a table, as shown in figure. A liquid film of thickness 0.25 mm is inserted between the block and the table. The block is pushed by a horizontal force of 0.1 N and moves with a constant speed. If the viscosity of the liquid is 5.0×10^{-3} Pl, the speed of block is _____ $\times 10^{-3}$ m/s.



Open solution

Q67. Numerical

A car is moving on a circular path of radius 600 m such that the magnitudes of the tangential acceleration and centripetal acceleration are equal. The time taken by the car to complete first quarter of revolution, if it is moving with an initial speed of 54 km/hr is $t(1 - e^{-\pi/2})s$. The value of t is _____.

Open solution

Q68. Numerical

Unpolarised light is incident on the boundary between two dielectric media, whose dielectric constants are 2.8 (medium -1) and 6.8 (medium -2), respectively. To satisfy the condition, so that the reflected and refracted rays are perpendicular to each other, the angle of incidence should be $\tan^{-1}\left(1 + \frac{10}{\theta}\right)^{\frac{1}{2}}$ the value of θ is _____.

(Given for dielectric media, $\mu_r = 1$)

Open solution

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Q69. Numerical

For a charged spherical ball, electrostatic potential inside the ball varies with r as $\mathrm{V} = 2ar^2 + b$.

Here, a and b are constant and r is the distance from the center. The volume charge density inside the ball is $-\lambda a\varepsilon$. The value of λ is _____.

ε = permittivity of the medium

Open solution

Q70. Numerical

When two resistance R_1 and R_2 connected in series and introduced into the left gap of a meter bridge and a resistance of 10 Ω is introduced into the right gap, a null point is found at 60 cm from left side. When R_1 and R_2 are connected in parallel and introduced into the left gap, a resistance of 3 Ω is introduced into the right gap to get null point at 40 cm from left end. The product of $\mathrm{R}_1\,\mathrm{R}_2$ is _____Ω^2

Open solution

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

A doctor prescribed the drug Equanil to a patient. The patient was likely to have symptoms of which disease?

(A) Stomach ulcers
(B) Depression and hypertension
(C) Anxiety and stress
(D) Hyperacidity

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

Match List I and List II

	List I		List II
A.	Osmosis	I.	Solvent molecules pass through semi permeable membrane towards solvent side.
B.	Reverse osmosis	II.	Movement of charged colloidal particles under the influence of applied electric potential towards oppositely charged electrodes.
C.	Electro osmosis	III.	Solvent molecules pass through semi permeable membrane towards solution side.
D.	Electrophoresis	IV.	Dispersion medium moves in an electric field.

Choose the correct answer from the options given below :

(A) A-III, B-I, C-IV, D-II
(B) A-III, B-I, C-II, D-IV
(C) A-I, B-III, C-IV, D-II
(D) A-I, B-III, C-II, D-IV

Open solution

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Q73. MCQ (Single Correct Answer) · **Chemistry**

The major component of which of the following ore is sulphide based mineral?

(A) Sphalerite (B) Siderite (C) Malachite (D) Calamine

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

The concentration of dissolved Oxygen in water for growth of fish should be more than $\underline{\mathrm{X}}$ ppm and Biochemical Oxygen Demand in clean water should be less than $\underline{\mathrm{Y}}$ ppm. X and Y in ppm are, respectively.

Open solution

(A)

X	Y
4	15

(B)

X	Y
4	8

(C)

X	Y
6	12

(D)

X	Y
6	5

Q75. MCQ (Single Correct Answer) · **Chemistry**

Match List I with List II

	List I		List II
A.	Elastomeric polymer	I.	Urea formaldehyde resin
B.	Fibre Polymer	II.	Polystyrene
C.	Thermosetting Polymer	III.	Polyester
D.	Thermoplastic Polymer	IV.	Neoprene

Open solution

Choose the correct answer from the options given below :

(A) A-II, B-I, C-IV, D-III (B) A-IV, B-I, C-III, D-II

(C) A-IV, B-III, C-I, D-II (D) A-II, B-III, C-I, D-IV

Q76. Numerical · **Chemistry**

A metal M forms hexagonal close-packed structure. The total number of voids in 0.02 mol of it is _____ $\times 10^{21}$ (Nearest integer).

(Given $\mathrm{N_A} = 6.02 \times 10^{23}$)

Open solution

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Q77. Numerical · **Chemistry**

On heating, $LiNO_3$ gives how many compounds among the following _____

$Li_2O, N_2, O_2, LiNO_2, NO_2$

Open solution

Q78. MCQ (More than One Correct Answer) · **Mathematics**

The statement $B \Rightarrow ((\sim A) \vee B)$ is equivalent to :

(A) $B \Rightarrow ((\sim A) \Rightarrow B)$

(B) $A \Rightarrow (A \Leftrightarrow B)$

(C) $A \Rightarrow ((\sim A) \Rightarrow B)$

(D) $B \Rightarrow (A \Rightarrow B)$

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

The plane $2x - y + z = 4$ intersects the line segment joining the points A $(a, -2, 4)$ and B $(2, b, -3)$ at the point C in the ratio 2 : 1 and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to :

(A) $\frac{17}{3}$

(B) $\frac{97}{3}$

(C) $\frac{16}{3}$

(D) $\frac{73}{3}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

If the tangent at a point P on the parabola $y^2 = 3x$ is parallel to the line $x + 2y = 1$ and the tangents at the points Q and R on the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ are perpendicular to the line $x - y = 2$, then the area of the triangle PQR is :

(A) $\frac{9}{\sqrt{5}}$

(B) $3\sqrt{5}$

(C) $5\sqrt{3}$

(D) $\frac{3}{2}\sqrt{5}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ and $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ intersect at the point P, then the distance of the point P from the plane $z = a$ is :

(A) 28

(B) 22

(C) 10

(D) 16

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The set of all values of $t \in \mathbb{R}$, for which the matrix

$$\begin{bmatrix} e^t & e^{-t}(\sin t - 2\cos t) & e^{-t}(-2\sin t - \cos t) \\ e^t & e^{-t}(2\sin t + \cos t) & e^{-t}(\sin t - 2\cos t) \\ e^t & e^{-t}\cos t & e^{-t}\sin t \end{bmatrix}$$ is invertible, is :

(A) $\{k\pi, k \in \mathbb{Z}\}$

(B) $\mathbb{R}$

(C) $\left\{(2k+1)\frac{\pi}{2}, k \in \mathbb{Z}\right\}$

(D) $\left\{k\pi + \frac{\pi}{4}, k \in \mathbb{Z}\right\}$

Open solution

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Q83. Numerical · **Mathematics**

A circle with centre (2, 3) and radius 4 intersects the line $x + y = 3$ at the points P and Q. If the tangents at P and Q intersect at the point $S(\alpha, \beta)$, then $4\alpha - 7\beta$ is equal to _____.

Open solution

Q84. Numerical · **Mathematics**

A triangle is formed by the tangents at the point (2, 2) on the curves $y^2 = 2x$ and $x^2 + y^2 = 4x$, and the line $x + y + 2 = 0$. If r is the radius of its circumcircle, then r^2 is equal to _____.

Open solution

Q85. Numerical · **Mathematics**

Let $a_1 = b_1 = 1$ and $a_n = a_{n-1} + (n-1), b_n = b_{n-1} + a_{n-1}, \forall n \geq 2$. If $S = \sum_{n=1}^{10} \frac{b_n}{2^n}$ and $T = \sum_{n=1}^{8} \frac{n}{2^{n-1}}$, then $2^7(2S - T)$ is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

If the equation of the normal to the curve $y = \frac{x-a}{(x+b)(x-2)}$ at the point (1, -3) is $x - 4y = 13$, then the value of $a + b$ is equal to _____.

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

With the help of potentiometer, we can determine the value of emf of a given cell. The sensitivity of the potentiometer is

(A) directly proportional to the length of the potentiometer wire

(B) directly proportional to the potential gradient of the wire

(C) inversely proportional to the potential gradient of the wire

(D) inversely proportional to the length of the potentiometer wire

Choose the correct option for the above statements :

(A) A only **(B)** B and D only **(C)** A and C only **(D)** C only

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

Substance A has atomic mass number 16 and half life of 1 day. Another substance B has atomic mass number 32 and half life of $\frac{1}{2}$ day. If both A and B simultaneously start undergo radio activity at the same time with initial mass 320 g each, how many total atoms of A and B combined would be left after 2 days.

(A) 1.69×10^{24} **(B)** 3.38×10^{24} **(C)** 6.76×10^{23} **(D)** 6.76×10^{24}

Open solution

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Q89. MCQ (Single Correct Answer) · **Physics**

The modulation index for an A.M. wave having maximum and minimum peak-to-peak voltages of 14 mV and 6 mV respectively is-

(A) 0.4 (B) 0.6 (C) 1.4 (D) 0.2

Open solution

Q90. Numerical · **Physics**

A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by 5Ω. When a resistance of 15Ω is used for shunting, null point moves to 300 cm. The internal resistance of the cell is _____Ω.

Open solution

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Answer Key

Chemistry

1. **C**	2. **A**	3. **B**	4. **D**	5. **A**
6. **C**	7. **D**	8. **C**	9. **D**	10. **D**
11. **B**	12. **D**	13. **D**	14. **C**	15. **B**
16. **15**	17. **17**	18. **4**	19. **4**	20. **200**
21. **4**	22. **270**	23. **2**		

Mathematics

24. **D**	25. **A**	26. **C**	27. **C**	28. **D**
29. **D**	30. **C**	31. **B**	32. **C**	33. **D**
34. **C**	35. **A**	36. **A**	37. **C**	38. **D**
39. **3000**	40. **603**	41. **14**	42. **9**	43. **9**
44. **5**				

Physics

45. **C**	46. **C**	47. **C**	48. **A**	49. **D**
50. **D**	51. **C**	52. **B**	53. **A**	54. **C**
55. **B**	56. **B**	57. **A**	58. **D**	59. **D**
60. **B**	61. **D**	62. **800**	63. **41**	64. **3872**
65. **40**	66. **25**	67. **40**	68. **7**	69. **12**
70. **30**				

Out of syllabus

71. **B**	72. **A**	73. **A**	74. **D**	75. **C**
76. **36**	77. **3**	78. **A, C, D**	79. **A**	80. **B**
81. **A**	82. **B**	83. **11**	84. **10**	85. **461**
86. **4**	87. **C**	88. **B**	89. **A**	90. **5**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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